

Continuous Celestial Attitude Estimation for Lunar Rover Motion via Relative Tracking and Catalog Re-Anchoring

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Abstract—We propose a continuous star-based attitude estimation framework for lunar exploration rovers operating in high-angular-velocity motion environments. Existing star-based absolute attitude estimation methods rely on star trackers, which require separate sensing hardware, or star catalog-based star identification; in both cases, performance can be degraded due to star streaks during high-angular-velocity motion, resulting in gaps in attitude estimation. To address these limitations, we estimate relative rotation from one frame to the next using learned correspondences between consecutive star images. Periodically, the estimated attitude is used to predict visible catalog stars and re-anchor the estimate. By combining relative rotation estimation with periodic re-anchoring, our method provides continuous and accurate absolute attitude estimates. We evaluate the proposed framework through real-world night-sky experiments using a single monochrome camera and rover-like motion profiles.

I. INTRODUCTION

Accurate attitude information is essential for the autonomous navigation and global positioning of lunar rovers. On planetary surfaces where external references such as GNSS and the Earth’s magnetic field are unavailable, star tracking provides a valuable source of absolute attitude. Dedicated star trackers have therefore been considered for planetary rover navigation, including direct attitude estimation [1] and periodic absolute updates for bounding rover-odometry drift [2]. However, they require dedicated sensing hardware and can become unreliable during rover motion [3].

Software-based Star Identification (Star-ID) methods instead recover absolute attitude from conventional camera images by matching observed star configurations to a catalog [4], [5], [6]. However, each absolute update still requires a clear enough star pattern in the current image to be matched reliably to the catalog. Image-based propagation reduces this dependence by estimating relative rotation between successive star images [7], but accumulates drift over time.

To address these limitations, we propose a continuous star-based attitude estimation system that combines learned frame-to-frame relative rotation tracking with quality-gated catalog re-anchoring. By avoiding a full Star-ID solve at

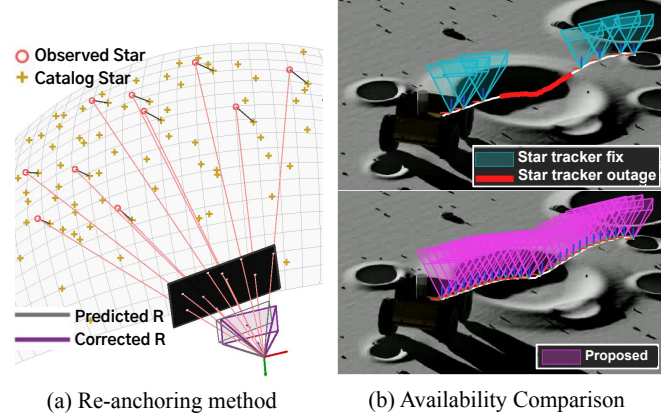


Fig. 1. Overview of the proposed framework. (a) Catalog re-anchoring of the propagated attitude using observed and projected catalog stars. (b) Output availability of a conventional star tracker and the proposed continuous tracking method.

every frame, our system supports high-angular-rate motion using only a single monochrome camera without gyroscopic aiding.

Our contributions are:

- We combine relative rotation tracking with periodic catalog re-anchoring to provide continuous, globally referenced attitude estimates.
- We enable high-angular-rate operation using a single monochrome camera without gyroscopic aiding.
- We validate the system under a real night sky using encoder ground truth and rover-like motion profiles.

II. PROPOSED METHOD

Fig. 1 illustrates the key idea of the proposed framework. The propagated attitude is corrected by aligning projected catalog stars with observed stars (Fig. 1(a)), while attitude tracking continues through temporary gaps in catalog-confirmed absolute fixes (Fig. 1(b)). As shown in Fig. 2(a), the pipeline takes a sequence of star images together with the calibrated camera model and star catalog as input, and performs a one-time Star-ID initialization to obtain the initial absolute attitude. For each subsequent frame, it estimates frame-to-frame relative rotation, predicts and re-anchors the attitude to the catalog, and updates the attitude estimate together with its validity.

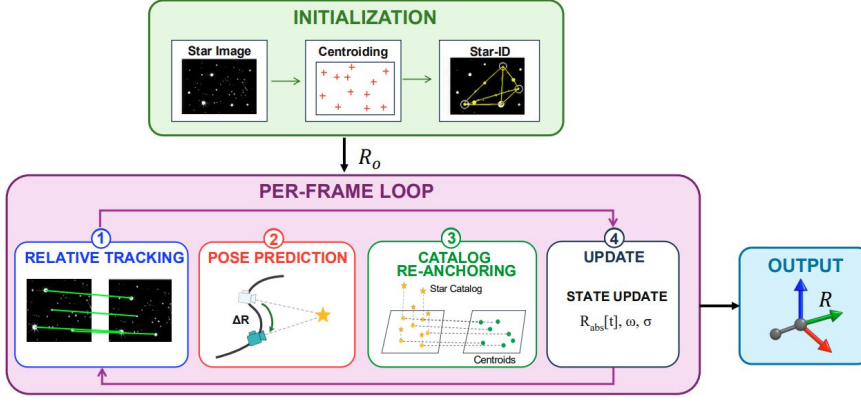
A. Initialization

For each input image, star centroids are first extracted using a standard thresholding and centroid computation method. A local background is estimated with a 51×51

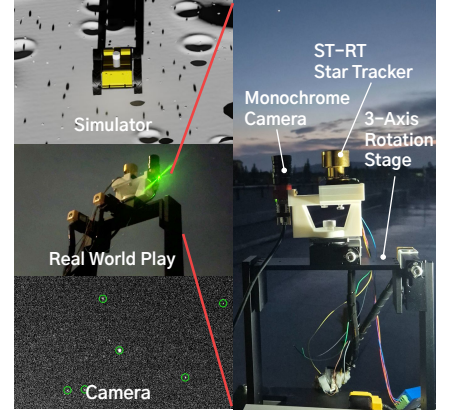
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(a) Overview



(b) Experimental Setup

Fig. 2. Proposed pipeline and experimental setup. (a) One-time initialization followed by the per-frame relative tracking, pose prediction, catalog re-anchoring, and state-update stages. (b) Motion-profile replay and data-acquisition setup using a monochrome camera, an ST-RT, and a 3-axis rotation stage.

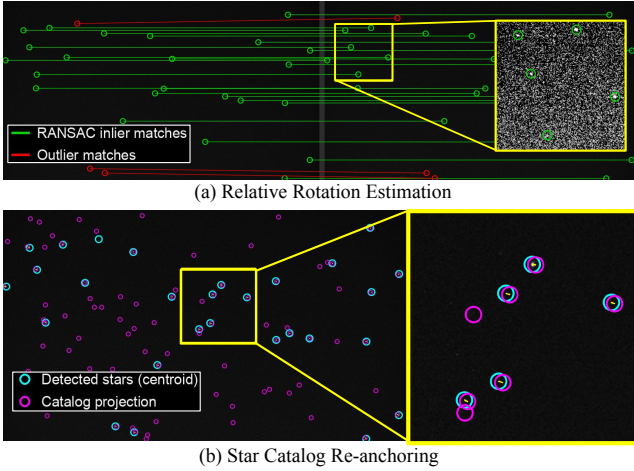


Fig. 3. Examples of the per-frame estimation stages. (a) Relative rotation estimation from RANSAC-filtered correspondences between consecutive star images (Intensity 10x). (b) Catalog re-anchoring by associating detected star centroids with projected catalog stars. Yellow line segments indicate the matched centroid-catalog pairs.

median filter and subtracted, and pixels exceeding 4σ of the per-frame sigma-clipped background noise are grouped by 8-connectivity; connected components of 4–200 pixels are kept, and each star’s sub-pixel position is taken as the intensity-weighted centroid of its component. The centroids are then undistorted with the calibrated intrinsics. Using centroiding information, the initial absolute attitude is obtained through Star-ID. We use Tetra [5] as the Star-ID method, which matches geometric patterns formed by 3–4 detected stars against a catalog to obtain $\mathbf{R}_0$.

Using this Star-ID solution as the initial absolute reference, the proposed system subsequently estimates the camera attitude $\mathbf{R}_t \in SO(3)$ for each frame t , where $\mathbf{v}_C = \mathbf{R}_t \mathbf{v}_S$ and $\mathbf{R}_t$ maps a vector from the celestial frame S to the camera frame C . This camera-frame convention is used throughout the absolute attitude representation.

Relative rotation estimation is internally performed using the OpenCV camera-coordinate convention. The fixed rota-

tion from the OpenCV frame O to the camera frame C is

$$\mathbf{P} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}, \quad (1)$$

such that $\mathbf{v}_C = \mathbf{P} \mathbf{v}_O$.

For a catalog star i with celestial unit vector $\mathbf{s}_i$, its direction in the camera frame is

$$[x_i, y_i, z_i]^\top = \mathbf{R}_t \mathbf{s}_i. \quad (2)$$

The corresponding image location is then obtained as

$$u_i = c_x - f_x \frac{y_i}{x_i}, \quad v_i = c_y - f_y \frac{z_i}{x_i}, \quad (3)$$

where only catalog stars within the forward camera field of view are retained. Since star sources are effectively at infinity, the projection depends only on camera rotation and does not include translation.

For relative tracking, detected centroids are first undistorted using the intrinsic parameters corresponding to the cropped image. Catalog projection uses $\mathbf{K}_{\text{tetra}}$, constructed from the field of view returned by the initial Star-ID solution.

B. Frame-to-Frame Propagation

After initialization, the system does not repeat a full Star-ID solve at every frame. Instead, it estimates relative rotation between consecutive images. As shown in Fig. 3(a), detected centroids in frames $t-1$ and t are matched using Light-Glue [8], which provides robust and efficient correspondence matching through its adaptive matching mechanism. Each image point is back-projected in the camera frame to a unit bearing vector,

$$\mathbf{b} = \frac{\mathbf{K}_c^{-1}[u, v, 1]^\top}{\|\mathbf{K}_c^{-1}[u, v, 1]^\top\|}. \quad (4)$$

The correspondences are processed using 500 RANSAC iterations, and the relative rotation is estimated by solving Wahba’s problem[9],

$$\hat{\mathbf{R}}_{\text{step}} = \arg \min_{\mathbf{R} \in SO(3)} \sum_{j \in \mathcal{I}} \|\mathbf{b}_j^t - \mathbf{R} \mathbf{b}_j^{t-1}\|^2, \quad (5)$$

where $\mathcal{I}$ denotes the RANSAC inlier set. Since $\hat{\mathbf{R}}_{\text{step}}$ is defined in the camera coordinates, the predicted absolute attitude is

$$\mathbf{R}_t^- = \mathbf{P} \hat{\mathbf{R}}_{\text{step}} \mathbf{P}^\top \mathbf{R}_{t-1}. \quad (6)$$

When relative tracking succeeds, the system updates a smoothed recent relative rotation $\mathbf{R}_v$. If matching or relative-rotation estimation fails, $\mathbf{R}_v$ is used to extrapolate the pose and keep the tracking state available for the next frame.

C. Catalog Re-Anchoring and State Update

Relative propagation can accumulate drift, so the system attempts catalog re-anchoring at every frame after initialization. The Yale Bright Star Catalogue 5 (BSC5) [10], also used for the initial Tetra Star-ID, is retained as a static celestial reference throughout tracking. Given the predicted attitude $\mathbf{R}_t^-$, catalog stars expected to be visible in the current field of view are projected onto the image and associated with the detected centroids, as shown in Fig. 3(b). These prediction-guided associations are used in a PnP-RANSAC/Wahba refinement to obtain a candidate absolute attitude $\mathbf{R}_t^{\text{ref}}$.

The candidate is accepted only after verifying its consistency with the catalog through reprojection. In our implementation, candidate associations are searched within 30 pixels, and the refined pose is validated using a 10-pixel reprojection gate; an uncertainty-based criterion is used when only a few stars are detected. If accepted, the state is updated as $\mathbf{R}_t = \mathbf{R}_t^{\text{ref}}$ and marked $\text{valid}_t = 1$; otherwise, the propagated attitude $\mathbf{R}_t^-$ is retained with $\text{valid}_t = 0$. Thus, the system maintains continuous attitude tracking while distinguishing catalog-confirmed absolute updates from propagated estimates.

The system also maintains a scalar uncertainty score σ_t . This quantity is a heuristic uncertainty measure rather than a calibrated covariance. It represents the increasing risk during propagation and is also used for the sparse-star acceptance condition. It is updated as

$$\sigma_t = \begin{cases} 0.2^\circ, & \text{re-anchoring accepted,} \\ \sigma_{t-1} + 0.05^\circ, & \text{rejected, relative tracking succeeds,} \\ \sigma_{t-1} + 0.3^\circ, & \text{rejected, relative tracking fails.} \end{cases} \quad (7)$$

For each frame, the system maintains the attitude state $\mathbf{R}_t$, the validity flag, σ_t , the relative-tracking status, and the re-anchoring status. This structure keeps the tracking state through image-based relative rotation without gyroscopic input or a vehicle-specific dynamics model, while catalog-based absolute updates are accepted only after the re-anchoring validation step.

III. EVALUATION

We evaluate the proposed pipeline using the real night-sky setup in Fig. 2(b). The experiments were conducted on the rooftop of GIST in Gwangju, Republic of Korea, under clear-sky conditions. A monochrome camera (Allied Vision

TABLE I
ATTITUDE RMSE OF THE PROPOSED PIPELINE AND STAR-TRACKER
AVAILABILITY

Rover	Sequence	RMSE [$^\circ$]	ST avail. [%]
Husky	A (waypoint)	0.269	81.1
	B (straight)	0.159	83.3
	C (long)	0.439	87.2
	D (zigzag)	0.313	87.7
Pragyaan	E (zigzag)	0.257	77.6
	F (straight, 0.2 m/s)	0.147	91.6

Alvium 1800 U-240m, 1936×1216) and a Rocket Lab ST-RT star Tracker are mounted on the same three-axis rotation stage, whose motor encoder provides the ground-truth attitude. The star tracker is used only for comparison and provides no input to the proposed pipeline. Thus, the method uses only a single monochrome camera without gyroscopic or other attitude sensing. Images are captured at 10fps with a $56.9^\circ \times 37.5^\circ$ field of view, a 100ms exposure, and a gain of 10. The star tracker is operated with an exposure time of 0.175s and an output rate of 1.1Hz. For repeatable testing, rover attitude trajectories from OmniLRS [11] are filtered, resampled to the 10Hz stage command rate, and replayed without angular scaling; profiles exceeding the motor rate limits are uniformly time-stretched. The stage encoders are logged at 100Hz and used as the attitude ground truth. We evaluate six sequences derived from two representative rover platforms: the four-wheeled Clearpath Husky ($0.99 \times 0.67 \times 0.39$ m) and the six-wheeled Pragyaan lunar rover ($0.917 \times 0.750 \times 0.397$ m). Sequences A–E are recorded at an average speed of 0.1 m/s, and sequence F at 0.2 m/s. Travel distances range from 2.6 to 13.9 m, with replay times of 50–189 s, covering different rover structures, paths, speeds, and terrain conditions.

Table I reports the attitude RMSE of the proposed pipeline against the motor-encoder GT and the output availability of the star tracker, obtained by counting frames in which the tracker fails to return a valid attitude. Across the six sequences, the RMSE ranges from 0.147° to 0.439° . The mean RMSE is 0.295° for Husky and 0.202° for Pragyaan. The star tracker availability ranges from 77.6% to 91.6%, corresponding to no valid attitude output for 8.4%–22.4% of each sequence. RMSE is computed only between the proposed method and the motor-encoder GT. Since the stage encoder is less accurate than the star tracker, we do not use it as a reference to evaluate the star tracker’s attitude accuracy.

Fig. 4 shows a top view of the trajectory and output availability for the representative sequence D. Fig. 4(a) shows the angular rate along the trajectory, with a maximum value of $14.2^\circ/\text{s}$. In Fig. 4(b), the star tracker fails to return a valid attitude for 12.3% of the sequence. The output gaps, shown in red, mainly occur near crater boundaries, where rapid changes in rover orientation lead to increased angular rates. In contrast, the proposed method in Fig. 4(c) maintains attitude estimates over the full trajectory under the same terrain and motion conditions. We also evaluate the proposed method with the frame-wise Star-ID methods Tetra,

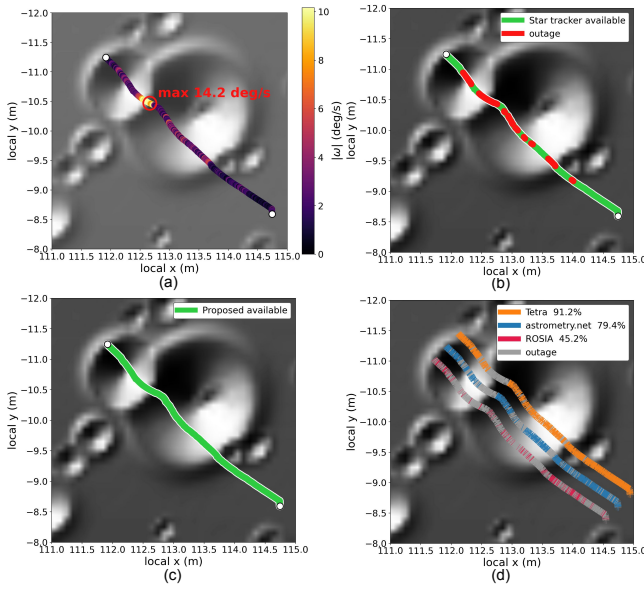


Fig. 4. Spatial output availability on Sequence D. (a) Rover trajectory colored by angular rate. (b) Star tracker output availability. (c) Proposed method output availability. (d) Output availability of frame-wise Star-ID methods.

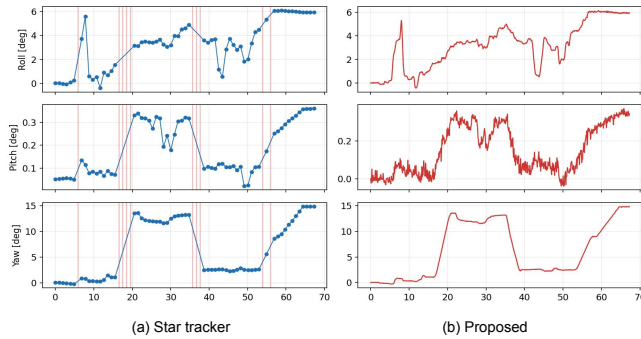


Fig. 5. Temporal attitude estimates over the same motion segment. (a) Star tracker outputs. (b) Proposed method outputs.

Astrometry.net [12], and ROSIA [6] on the same sequence. Astrometry.net is a blind plate solver based on a prebuilt geometric index, whereas ROSIA searches directly over the rotation space. As shown in Fig. 4(d), their output availability values are 91.2% for Tetra, 79.4% for Astrometry.net, and 45.2% for ROSIA. These methods estimate the absolute attitude independently from each image and therefore show failure gaps during fast-motion intervals. Our method instead propagates the attitude using relative rotation between consecutive frames and re-anchors it to the catalog when needed. It therefore maintains continuous attitude estimates over the full sequence, including motion up to $14.2^\circ/\text{s}$.

Fig. 5 compares the temporal changes in roll, pitch, and yaw produced by the star tracker and the proposed method over the same representative segment. In Fig. 5(a), the star tracker returns 61 valid fixes and fails to solve 10 times. In contrast, the proposed method in Fig. 5(b) produces attitude estimates continuously for 669 frames over the same interval. This corresponds to an attitude stream approximately 11 times denser than the valid star tracker fixes. The star tracker failures mainly occur during rapid changes, where fast mo-

tion causes star streaks and degrades centroid estimation. The proposed method maintains its estimate through these intervals, while the overall trends of all three attitude axes remain similar at the times when valid star tracker outputs are available.

IV. CONCLUSION

We propose a single-camera attitude-estimation framework that maintains star-based attitude tracking during fast lunar rover motion by combining initial Star-ID, frame-to-frame relative rotation, and catalog re-anchoring. The system maintains the attitude state without per-frame Star-ID, gyroscopic input, or a vehicle-specific dynamics model. In experiments using real night-sky images and rover-like motion profiles, the proposed method achieves a maximum attitude RMSE of 0.439° . The current Earth-based night-sky experiments differ from lunar observing conditions and are limited by the rotation-stage reference accuracy. Future work will focus on validation under lunar-like conditions and improving robustness to initial Star-ID failure.

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